

Question	Answer	Mark	Mark scheme	Additional guidance
15 (a)	$\overrightarrow{AC} = 5\overrightarrow{AB}$ and reason	M1	for $\overrightarrow{AC} = 5(3\mathbf{a} + 4\mathbf{b})$ or $\overrightarrow{BC} = 4(3\mathbf{a} + 4\mathbf{b})$ or indicates that $15\mathbf{a} + 20\mathbf{b} = 5 \times (3\mathbf{a} + 4\mathbf{b})$	Do not award this mark if any incorrect working seen Could use AB and BC or AC and BC
		C1	for $\overrightarrow{AC} = 5\overrightarrow{AB}$ or $\overrightarrow{AC} = 5(3\mathbf{a} + 4\mathbf{b})$ or $15\mathbf{a} + 20\mathbf{b} = 5(3\mathbf{a} + 4\mathbf{b})$ and a correct reason, eg AC is a multiple of AB / multiples of each other / AB is a factor of AC or they have the same gradient / are parallel / go in the same direction or they have a point in common / both start at A or AC is an enlargement of AB	
(b)	5 : 2	P1	for ($\overrightarrow{DF} =$) $3\mathbf{e} + 6\mathbf{f} + (-10.5\mathbf{e} - 21\mathbf{f})$ ($= -7.5\mathbf{e} - 15\mathbf{f}$) or for a multiplicative relationship for $\overrightarrow{DE}$ and $\overrightarrow{EF}$ eg $\overrightarrow{EF} = -3.5 \overrightarrow{DE}$ or for $(DE : EF =)$ $1 : -3.5$ oe	
		P1	for a multiplicative relationship for $\overrightarrow{DF}$ and $\overrightarrow{DE}$ eg $\overrightarrow{DF} = -2.5 \overrightarrow{DE}$ or for $(DF : DE =)$ $-5 : 2$ oe eg $-2.5 : 1$ or $-7.5 : 3$ or for answer of $2 : 5$ oe	
		A1	oe eg $2.5 : 1$	